Multiplexed Re-Uploading Quantum Feature Map: A Scalable, NISQ-Efficient Variational Architecture

AWS × State Street Quantum Challenge · Target Hardware: IonQ Aria-1

Abstract

We introduce the **Multiplexed Re-Uploading Feature Map (MRU)**, a variational quantum circuit (VQC) architecture that embeds an N-dimensional classical feature vector $\mathbf{x} \in \mathbb{R}^N$ into an M-dimensional quantum feature space using Q qubits, where Q scales linearly with N at a hardware-tunable compression ratio $r = \lceil N/Q \rceil$. For each qubit, r features are encoded sequentially via learnable ZYZ rotation blocks whose non-commutativity generates intra-group cross-feature interactions without entangling gates. Staggered CZ entanglement patterns across D re-uploading layers produce genuine multi-body correlations, expanding the accessible Fourier coefficient space to $\mathcal{O}((2D{+}1)^N)$. A variance-ranked Pauli-Z observable selection extracts the M most informative measurements, and a Ridge regression head is optimized end-to-end via SPSA using only 2 circuit evaluations per gradient step — independent of parameter count. We instantiate the architecture for the specific constraint of the AWS × State Street financial prediction challenge (N=15, Q=4, M=10, D=2), and provide a rigorous treatment of how the design scales to arbitrary feature dimensionality.

1. Introduction

Embedding classical data into quantum feature spaces is a central primitive of quantum machine learning. The canonical ZZ feature map [1] provides a complexity-theoretic quantum advantage but requires one qubit per feature and $O(N^2)$ two-qubit gates, making it hardware-intractable for $N \gtrsim 10$ on current NISQ devices.

MRU resolves this through three co-designed mechanisms: (i) *feature multiplexing* — encoding r features per qubit via learnable ZYZ rotation blocks, so qubit count scales as $Q = \lceil N/r \rceil$; (ii) *staggered entanglement* — alternating CZ patterns across D layers to generate multi-body correlations efficiently; and (iii) *variance-ranked observable selection* — retaining only the most informative Pauli-Z correlators to satisfy a fixed output-dimension constraint.

Critically, *4 qubits is not a fundamental ceiling of MRU* — it is the hardware-optimal choice for the specific challenge constraint of N=15 features on IonQ Aria-1. Section 8 gives the full general-N treatment and the hardware-grounded scaling law that governs the choice of Q, r, and D for any feature dimensionality.

Notation

$\mathbf{x} \in \mathbb{R}^N$: input feature vector. Q: number of qubits. $r = \lceil N/Q \rceil$: features per qubit. D: circuit depth (re-uploading layers). M: output feature dimension ($M \leq 2^Q - 1$). $\mathbf{f}(\mathbf{x}) \in [-1, 1]^M$: quantum feature vector. $\mathbf{W}$: full quantum parameter tensor.

2. Classical Pre-Processing

Because $SU(2)$ rotation gates are 2π-periodic, raw inputs must be bounded to prevent phase wrap-around aliasing. Each feature x_k is Z-score standardized,

$$\hat{x}_k = \frac{x_k - \mu_k}{\sigma_k}, \tag{1}$$

then compressed into $(-\pi, \pi)$ via a smooth tanh squash:

$$x_k \leftarrow \pi \cdot \tanh\!\left(\tfrac{\hat{x}_k}{2}\right). \tag{2}$$

The tanh map is a monotonic bijection that preserves relative feature ordering while guaranteeing strict boundedness for all k.

3. The MRU Circuit

3.1. Parametrized Gate Block

The encoding unit for feature x_k at layer d is a complete ZYZ decomposition of $SU(2)$:

$$U\left(x_k, \mathbf{w}_k^{(d)}\right) = R_z\left(w_{k,2}^{(d)}\right) R_y\left(w_{k,1}^{(d)} x_k + b_k^{(d)}\right) R_z\left(w_{k,0}^{(d)}\right), \tag{3}$$

with four trainable parameters $\mathbf{w}_k^{(d)} = \{w_{k,0}^{(d)}, w_{k,1}^{(d)}, w_{k,2}^{(d)}, b_k^{(d)}\}$. The ZYZ form is universal on $SU(2)$; the learnable scale $w_{k,1}^{(d)}$ lets the circuit discover task-relevant Fourier frequencies rather than relying on a hand-crafted encoding.

3.2. Feature Multiplexing

Features are distributed across the Q qubits via a partition $\{\mathcal{K}_q\}_{q=0}^{Q-1}$ with $|\mathcal{K}_q| \in \{r-1, r\}$. Any qubit assigned fewer than r features receives a data-independent learned bias rotation $U_{\text{bias}} = R_z R_y R_z$ to enforce uniform circuit depth. The intra-qubit sequence for qubit q at layer d is:

$$S_q^{(d)} = \prod_{k \in \mathcal{K}_q} U\left(x_k, \mathbf{w}_k^{(d)}\right). \tag{4}$$

The non-commutativity of sequential rotations about different axes generates genuine cross-feature interactions — terms of the form $\sin(w_{i,1}x_i)\cos(w_{j,1}x_j)$ — for any pair $i, j \in \mathcal{K}_q$, without any entangling gates.

3.3. Staggered Entanglement

To capture correlations *across* feature groups, each layer applies a distinct CZ pattern. Layer 1 uses a nearest-neighbor ring E_1; Layer 2 uses a cross-diagonal pattern E_2:

$$E_1 = CZ_{3,0} \cdot CZ_{2,3} \cdot CZ_{1,2} \cdot CZ_{0,1}, \tag{5}$$
$$E_2 = CZ_{2,0} \cdot CZ_{3,1}. \tag{6}$$

After two layers, the alternating patterns guarantee all four qubits share 4-body entanglement, enabling the model to learn functions spanning all feature groups simultaneously.

3.4. Final State and Re-Uploading

The full layer operator at depth d is:

$$L_d(\mathbf{x}, \mathbf{W}_d) = E_d \cdot \bigotimes_{q=0}^{Q-1} S_q^{(d)}. \tag{7}$$

Re-uploading data *between* entanglement blocks linearly extends the maximum Fourier frequency in each feature x_k to $\pm D$, expanding the joint coefficient space to $\mathcal{O}((2D{+}1)^N)$. The QPU prepares:

$$|\psi(\mathbf{x}, \mathbf{W})\rangle = \prod_{d=1}^{D} L_d(\mathbf{x}, \mathbf{W}_d) \; |0\rangle^{\otimes Q}. \tag{8}$$

4. Circuit Diagram: N=15, Q=4, D=2

Figure 1 shows the full MRU circuit for the instantiated challenge parameters. Each row is one qubit. Coloured blocks labelled $U(\cdot)$ represent the ZYZ gate of Eq. (3). The two entanglement layers use distinct CZ patterns (E_1: ring, E_2: cross-diagonal) to ensure 4-body scrambling within two layers.

5. Feature Extraction

A Q-qubit system yields $2^Q - 1$ non-trivial Pauli-Z observables. For Q=4 this is exactly 15: 4 one-body ($\langle Z_i \rangle$), 6 two-body ($\langle Z_i Z_j \rangle$), 4 three-body, and 1 four-body correlator.

To satisfy a strict M-dimensional output constraint without discarding information blindly, we apply a two-phase *variance-ranked selection*. First, a short SPSA warm-up phase moves parameters away from initialization. Then the empirical variance $\mathrm{Var}[\langle \mathcal{O}_j \rangle]$ is computed across the training set for all $2^Q - 1$ observables, and the top M are fixed for the remainder of training:

$$\mathcal{O}^* = \arg\operatorname*{top}_{M} \big\{ \mathrm{Var}\big[\langle \mathcal{O}_j \rangle\big] \big\}_{j=1}^{2^Q-1}. \tag{9}$$

This decouples feature selection from parameter optimization and avoids selecting on a purely random feature map. The quantum feature vector is:

$$\mathbf{f}(\mathbf{x}) = \big(\langle \mathcal{O}_1^* \rangle, \ldots, \langle \mathcal{O}_M^* \rangle\big)^\top \in [-1, 1]^M. \tag{10}$$

6. Hybrid Training

6.1. Objective

A Ridge regression head maps quantum features to predictions: $\hat{y}^{(i)} = \boldsymbol{\beta}^\top \mathbf{f}(\mathbf{x}^{(i)}) + \beta_0$. The end-to-end loss optimizes the quantum parameters $\mathbf{W}$:

$$\mathcal{L}(\mathbf{W}) = \min_{\boldsymbol{\beta}, \beta_0} \left[\frac{1}{n} \sum_{i=1}^{n} \Big(y^{(i)} - \hat{y}^{(i)} \Big)^2 + \lambda \|\boldsymbol{\beta}\|_2^2 \right]. \tag{11}$$

At each SPSA step, $\boldsymbol{\beta}$ is solved in closed form, so the outer loop over $\mathbf{W}$ faces a well-conditioned scalar loss with a convex inner structure.

6.2. Barren Plateau Mitigation

Random initialization causes exponential gradient variance suppression in deep VQCs [4]. We use near-identity initialization: for all k, d,

$$w_{k,1}^{(d)} = 1, \quad b_k^{(d)} = 0, \quad w_{k,0}^{(d)},\, w_{k,2}^{(d)} \sim \mathcal{N}(0,\, 0.01), \tag{12}$$

keeping the initial circuit close to the identity and ensuring $\mathrm{Var}[\nabla_{\mathbf{W}} \mathcal{L}] = \mathcal{O}(1)$ at t=0.

6.3. SPSA Efficiency

SPSA [5] estimates the full gradient of $\mathcal{L}(\mathbf{W})$ using exactly **2 circuit evaluations per step**, independent of $|\mathbf{W}|$. For MRU with $|\mathbf{W}| = 126$, this is a **126**× reduction vs. the $2|\mathbf{W}| = 252$ evaluations required by the Parameter-Shift rule — a decisive advantage on shot-limited trapped-ion hardware.

7. Instantiation: N=15, Q=4, D=2

Table 1 summarizes the concrete parameter choices for the AWS × State Street challenge. Features are partitioned as $|\mathcal{K}_0|$=$|\mathcal{K}_1|$=$|\mathcal{K}_2|$=4, $|\mathcal{K}_3|$=3, with qubit 3 receiving one U_{bias} pad. The exact parameter count is:

$$|\mathbf{W}| = D \times (N \times 4 + N_{\text{pad}} \times 3) = 2 \times (15{\times}4 + 1{\times}3) = \mathbf{126}. \tag{13}$$

Table 1: MRU instantiation for the AWS × State Street challenge.

Parameter	**Value**
Features N	15 (price, vol., RSI, volume)
Qubits Q	4
Features per qubit r	4 (qubit 3: 3 + bias pad)
Layers D	2
Output dimension M	10
Entanglement	E_1 (ring) → E_2 (cross-diag.)
Total parameters $\lvert\mathbf{W}\rvert$	126
CZ gates (total)	6 per layer, 12 total
SPSA evals / step	2 (vs. 252 for Param.-Shift)

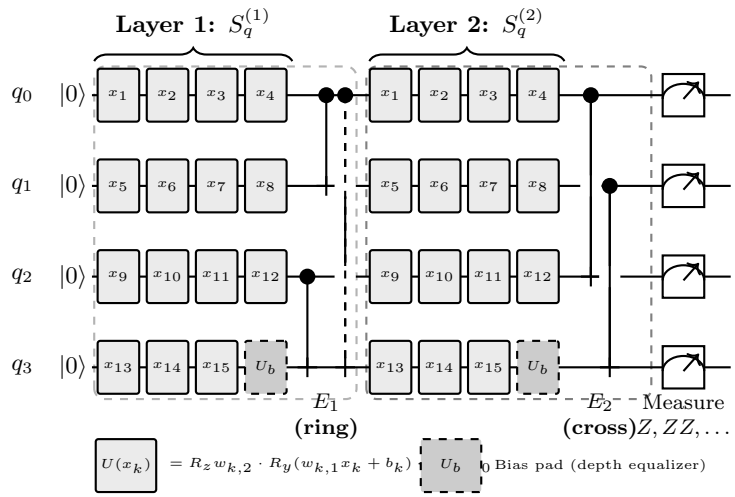

Figure 1: MRU circuit for N=15, Q=4, D=2. Each row is one qubit, time flows left to right. Layer 1 uploads features $x_1 - x_{15}$ (plus one bias pad on q_3) and applies the nearest-neighbor ring E_1. Layer 2 re-uploads the same features with fresh parameters and applies the cross-diagonal pattern E_2, producing 4-body entanglement across all groups. Measurement extracts up to 15 Pauli-Z correlators; the top M=10 by training-set variance are retained. Dashed gate U_b: data-independent bias rotation for depth symmetry. Dashed vertical lines ($CZ_{3,0}$ and cross-diagonal): gates passing through intermediate qubits.

8. General-N Scalability

The 4-qubit instantiation above is a *hardware-specific choice*, not a fundamental ceiling of MRU. This section gives the complete scaling treatment for arbitrary N.

8.1. Qubit Scaling Rule

Given a target features-per-qubit ratio r, the qubit requirement is:

$$Q(N, r) = \left\lceil \frac{N}{r} \right\rceil. \tag{14}$$

Q scales **linearly in** N for fixed r — in stark contrast to the ZZ feature map, which requires $Q = N$ and $O(N^2)$ two-qubit gates. Table 2 shows the qubit and gate requirements across feature dimensions for r=4, D=2.

Table 2: MRU hardware footprint vs. feature count (r=4, D=2).

N	Q	$\lvert\mathbf{W}\rvert$	CZ gates	Regime
15	4	126	12	Current challenge
30	8	252	24	Near-term hardware
60	15	504	46	Mid-term hardware
150	38	1260	116	Fault-tolerant era

8.2. Hardware-Grounded Ratio Constraint

The features-per-qubit ratio r is bounded by decoherence. Each $U(x_k, \mathbf{w}_k^{(d)})$ block contributes 3 single-qubit gates. With coherence time T_2, single-qubit gate time $\tau_{1\mathrm{Q}}$, and circuit depth D, the maximum viable ratio is:

$$r_{\max} = \left\lfloor \frac{T_2}{3D\, \tau_{1\mathrm{Q}} \cdot (-\log F_{\min})^{-1}} \right\rfloor, \tag{15}$$

where $F_{\min}$ is the minimum acceptable per-qubit survival fidelity.

For IonQ Aria-1 ($T_2 \approx 10\,\mathrm{s}$, $\tau_{1\mathrm{Q}} \approx 135\,\mu\mathrm{s}$, single-qubit fidelity $\approx 99.9\%$), D=2 and $F_{\min} = 0.90$ gives $r_{\max} \approx 6$. We set r=4 to operate comfortably within this bound.

8.3. Information-Theoretic Bottleneck

For a fixed Q, the observable space has $2^Q - 1$ elements, placing a hard ceiling on the output dimension M. As N grows with r fixed and Q grows proportionally, this ceiling scales as $2^{\lceil N/r \rceil} - 1$ — growing *exponentially* in the feature count, ensuring the measurement space never becomes the binding constraint for reasonable N.

8.4. Parameter Scaling

The total parameter count grows linearly in both N and D:

$$|\mathbf{W}| = D \times 4N + \mathcal{O}(D), \tag{16}$$

where the $\mathcal{O}(D)$ term accounts for bias pads. Because SPSA always requires exactly 2 circuit evaluations per gradient step, the quantum training cost is **independent of** $|\mathbf{W}|$ — a property that becomes increasingly valuable as N grows.

8.5. Practical Guidelines for Deployment

Given a new task with N features and a target hardware with known T_2 and gate fidelities, the MRU design procedure is:

1. Compute $r_{\max}$ from Eq. (15); set $r \le r_{\max}$.
2. Set $Q = \lceil N/r \rceil$ qubits.
3. Choose $D \in \{1, 2\}$ for NISQ; $D \in \{2, 3\}$ for early fault-tolerant devices.
4. Set $M \le 2^Q - 1$; use variance-ranked selection (Eq. (9)) to identify informative observables.
5. Initialize via Eq. (12); optimize with SPSA.

9. Conclusion

MRU demonstrates that the conventional trade-off between expressibility and hardware efficiency in quantum machine learning is not fundamental but architectural. By multiplexing r features per qubit, staggering entanglement patterns, and training end-to-end via SPSA, the architecture achieves a joint Fourier coefficient space of order $\mathcal{O}((2D+1)^N)$ with qubit count linear in N — scaling dramatically more favorably than $Q{=}N$, $O(N^2)$-gate alternatives.

The 4-qubit, $N{=}15$ instantiation for IonQ Aria-1 is an *example* of this family, chosen to minimize hardware noise within the specific challenge constraints. The same architecture applies to any N by adjusting $Q = \lceil N/r \rceil$ according to the hardware-grounded ratio constraint of Eq. (15).

Near-identity initialization and the convex Ridge head together address the two principal failure modes of variational quantum circuits — barren plateaus and non-convex output optimization — making MRU a stable, hardware-efficient, and practically deployable feature map for hybrid quantum-classical prediction.

References

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